

FANR 3800: Week 2 Student Notes

Spatial Analysis for Natural Resources · Fall 2026

Invalid Date

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1 Week 2 - Telling the Spatial Story

Name: _____ Date: _____

1.1 How to Use This Sheet: Two-Pass Notes

In class, capture the key idea under each prompt in your own words: a phrase or a sentence, not everything. That evening, reread your notes and use the “After class: add detail” box at

the bottom of each page to add detail from the assigned ESRI Training exercise and the eLC links. Need more room? Use your own paper.

1.2 1 · This Week (10 min)

Weekly schedule, ESRI Training, deliverables, and due dates.

1.3 2 · Lab Review: Last Week's Setup (10 min)

What did getting ArcGIS Pro set up show me about the project and interface? Anything still not working?

Why does having professional GIS software matter for someone in my discipline?

1.4 3 · The Week Ahead (5 min)

New ideas this week: *Symbology* (how we show features) and *Map Layout* (what makes a map credible).

1.5 4 · Application Discussion (10 min) (may vary; instructor's call)

Read the prompt for **your** focus and jot a response; be ready to share.

- **Forestry/Forest Management:** A forester maps timber stand types for a landowner. How does the choice of *Symbology* (colors, patterns) affect a non-GIS-savvy landowner's understanding?
- **Wildlife Management:** Mapping critical red-cockaded woodpecker habitat in a national forest, which essential map elements would you prioritize so managers can interpret it quickly?
- **Fisheries Management:** Mapping fish sampling locations in a reservoir, how can map design distinguish species caught or water-quality values for a non-technical audience?

- **Parks Recreation & Tourism Management:** How would a visitor map's purpose (guidance) change your design choices compared with an internal maintenance map?

1.6 5 · Lecture: *Symbology*, Making Attributes Visible (15 min)

- Symbology is *analysis*, not decoration: the same data with different symbols can tell different stories.

- The main methods: *Single Symbol*; *Categorized / Unique Values* (qualitative, e.g., species, ownership); *Graduated* (quantitative, e.g., age, basal area) and how class breaks change the message.

- Choosing well: color for categories vs. sequences; how many classes; why raw counts can mislead (normalize instead).

Key terms: *Symbology* · *Categorized* · *Graduated* · *Unique Values* · class breaks

1.7 After Class: Add Detail

From the assigned ESRI Training exercise and the eLC links, add which symbology method best fits a dataset in *your* field, and why.

2 WEDNESDAY

2.1 1 · Lecture: *Map Layout, the Anatomy of a Credible Map (20 min)*

- The essential elements (title, legend, scale bar, north arrow, and data source plus coordinate-system metadata), and what a *missing* element costs the reader's trust.
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- Composition and visual hierarchy: guiding the eye; a map for a report page vs. a map for a wall.
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- Credibility and honesty: how framing, extent, and class breaks can distort; cite sources; show uncertainty.
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- Export and delivery: EXPORT LAYOUT; format and resolution for print vs. screen; consistent file naming.
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Key terms: legend · scale bar · north arrow · metadata · visual hierarchy · EXPORT LAYOUT

2.2 2 · Pre-lab Review: *Why This Week's Lab Matters (10 min)*

This week's lab has you build and symbolize a map. Why is a clear, credible map a professional deliverable in my field?

2.3 3 · Pre-lab Review: *Methods & Tools (10 min)*

The tools and steps I will use this afternoon: the SYMBOLOGY pane, the LAYOUT view, and EXPORT LAYOUT.

2.4 4 · Week Summary (10 min)

In your own words, the one thing to remember from Week 2:

Questions I still have (bring to Monday):

Deliverables: ESRI Training certificate due Wed 8:40 AM; lab document due Sunday 11:59 PM.

2.5 After Class: Add Detail

From the assigned ESRI Training exercise and the eLC links, write the checklist of elements you will put on *every* map, and add the one you are most likely to forget.

2.6 Evening Review Checklist (Second Pass)

- ☐ Reread my in-class notes within 24 hours
- ☐ Added detail from the assigned ESRI Training exercise
- ☐ Added detail from the eLC web links
- ☐ Wrote my own one-sentence summary of the week
- ☐ Listed questions to ask on Monday